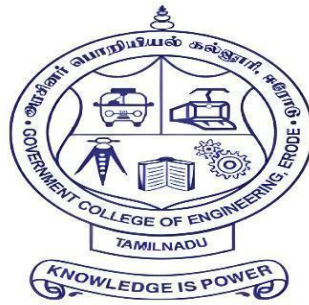


**GOVERNMENT COLLEGE OF ENGINEERING (IRTT)
ERODE-638316**



PASSWORD-BASED SMART LOCK SYSTEM

A PROJECT REPORT

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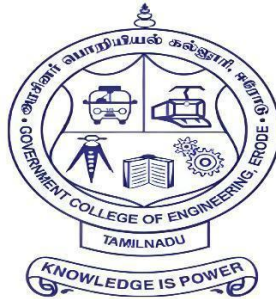
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BONAFIDE CERTIFICATE



Certified that this project report titled “Password -based
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TABLE OF CONTENTS

S.NO	TITLE	PAGE NO
	Abstract	5
1	Introduction	6
2	Objectives	7
3	Hardware Requirements	7
4	Software Requirements	9
5	Methodology	9
6	System Architecture	10
7	Working Principle	11
8	Project Code	11
9	Results and Discussion	13
10	Future Enhancements	14
11	Conclusion	15

ABSTRACT

In the modern world, ensuring security has become a major concern in residential, commercial, and industrial environments. Traditional locking systems that rely on physical keys are vulnerable to various security issues such as key duplication, loss, and unauthorized access. To overcome these limitations, a Password-Based Smart Lock System using Raspberry Pi is proposed.

This system uses a keypad module to allow users to enter a password. The entered password is processed by the Raspberry Pi and compared with a pre-stored password. If the password matches, the system grants access by activating a relay that unlocks the door. If the password is incorrect, access is denied, thereby ensuring security.

The system is cost-effective, easy to implement, and highly reliable. It eliminates the need for physical keys and enhances security using embedded systems and automation. This project demonstrates how simple electronic components can be integrated to create an efficient and smart access control system suitable for homes, offices, and restricted areas.

1.Introduction

Security plays a vital role in today's rapidly developing world. With increasing concerns about theft and unauthorized access, there is a need for smarter and more reliable security systems. Traditional locking mechanisms, which depend on mechanical keys, are prone to several disadvantages such as key duplication, misplacement, and lack of monitoring.

To address these issues, electronic locking systems have been developed. One such system is the password-based smart lock system, which replaces traditional keys with a secure password entry mechanism. In this system, a keypad is used to input a password, and a microcontroller or embedded system verifies the input.

This project uses Raspberry Pi as the core controller to implement a smart locking mechanism. The system checks whether the entered password matches the stored password. If it matches, access is granted; otherwise, access is denied.

This system is simple, efficient, and cost-effective. It is widely applicable in homes, offices, lockers, and restricted areas where security is essential.

2.Objectives

The main objectives of this project are:

- To design and develop a password-based smart lock system
- To implement secure access control using Raspberry Pi
- To replace traditional key-based systems with digital authentication
- To allow access only to authorized users
- To improve security and reliability
- To develop a low-cost and user-friendly system

3.Hardware Requirements

1.Raspberry Pi 5 :

- ☐ Acts as the main controller
- ☐ Processes keypad input
- ☐ Controls the relay and locking mechanism
- ☐ Runs Python programs for authentication



2.4x4 Keypad Module

- Used for entering password
- Contains 16 keys (0–9, A–D, *, #)
- Sends input signals to Raspberry Pi

3. MicroSD Card (64GB):

Used to store the operating system and datasets required for the system.



4. Power Supply (USB-C Adapter):

Provides the necessary power to operate the Raspberry Pi.

5. Relay Module

- Acts as an electronic switch
- Controls the door lock mechanism
- Operates based on Raspberry Pi output.



6. Solenoid Door Lock / DC Motor

- Provides physical locking and unlocking
- Activated through relay



4. Software Requirements

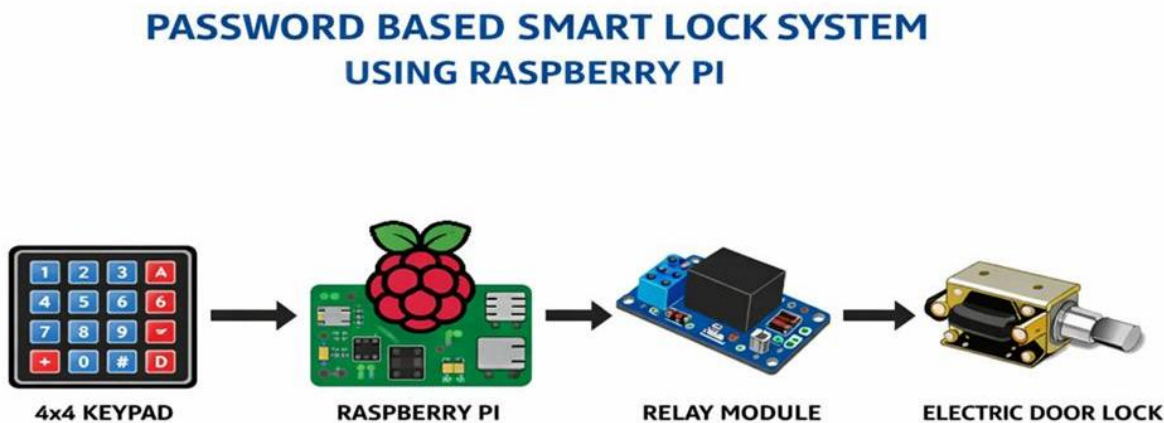
- **Python** – Programming language used
- **Raspberry Pi OS** – Operating system
- **PuTTY / RealVNC Connect**- For coding and execution
- **RPi.GPIO Library** – For controlling GPIO pins

5. Methodology

The working of the system is divided into several steps:

1. The system initializes Raspberry Pi and keypad connections
2. The user enters the password using the keypad
3. Each key press is read and stored temporarily
4. When the user presses the “Enter key (#)”, the system verifies the password
5. The entered password is compared with the stored password
6. If the password matches:
 - Access is granted
 - Relay is activated
 - Door unlocks
7. If the password does not match:
 - Access is denied
 - Door remains locked
8. The system resets and waits for the next input

6. System Architecture



7. Working Principle

The system works based on password authentication.

- The keypad is interfaced with Raspberry Pi GPIO pins
- User enters password through keypad
- Raspberry Pi reads the input using Python program
- Input password is compared with stored password
- If correct:
 - Relay turns ON
 - Door unlocks
- If incorrect:
 - Relay remains OFF
 - Door stays locked

The system operates continuously and provides real-time access control.

8. Project Code

```
import RPi.GPIO as GPIO
import time
password = "1234"
input_password = ""
GPIO.setmode(GPIO.BCM)
rows = [5, 6, 13, 19]
cols = [12, 16, 20, 21]
for row in rows:
    GPIO.setup(row, GPIO.OUT)
    GPIO.output(row, 1)
for col in cols:
    GPIO.setup(col, GPIO.IN, pull_up_down=GPIO.PUD_UP)
```

```

keys = [
['1','2','3','A'],
['4','5','6','B'],
['7','8','9','C'],
['*','0','#','D']
]
def read_key():
for i, row in enumerate(rows):
GPIO.output(row, 0)
for j, col in enumerate(cols):
if GPIO.input(col) == 0:
time.sleep(0.2)
GPIO.output(row, 1)
return keys[i][j]
GPIO.output(row, 1)
return None
try:
while True:
key = read_key()
if key:
print("Pressed:", key)
if key == '#':
if input_password == password:
print("UNLOCKED")
else:
print("WRONG PASSWORD")
input_password = ""
elif key == '*':
input_password = ""
else:
input_password += key
except KeyboardInterrupt:
GPIO.cleanup()

```

FOR WEB DEVELOPMENT

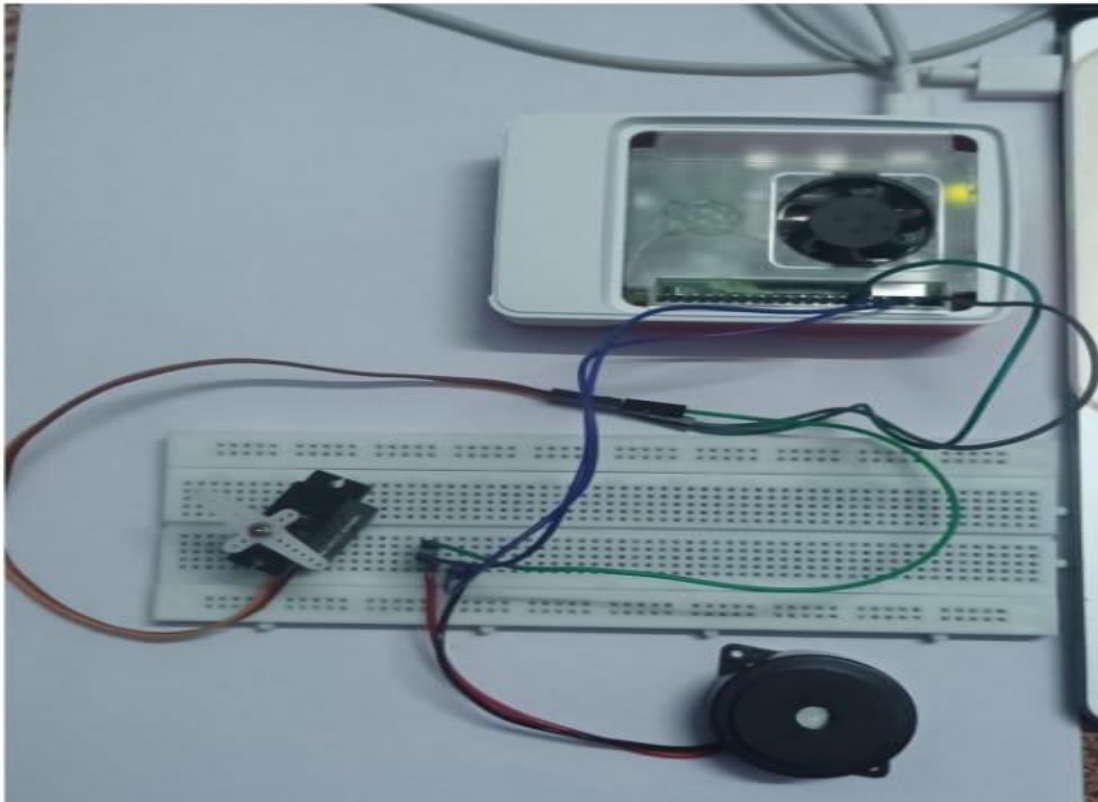
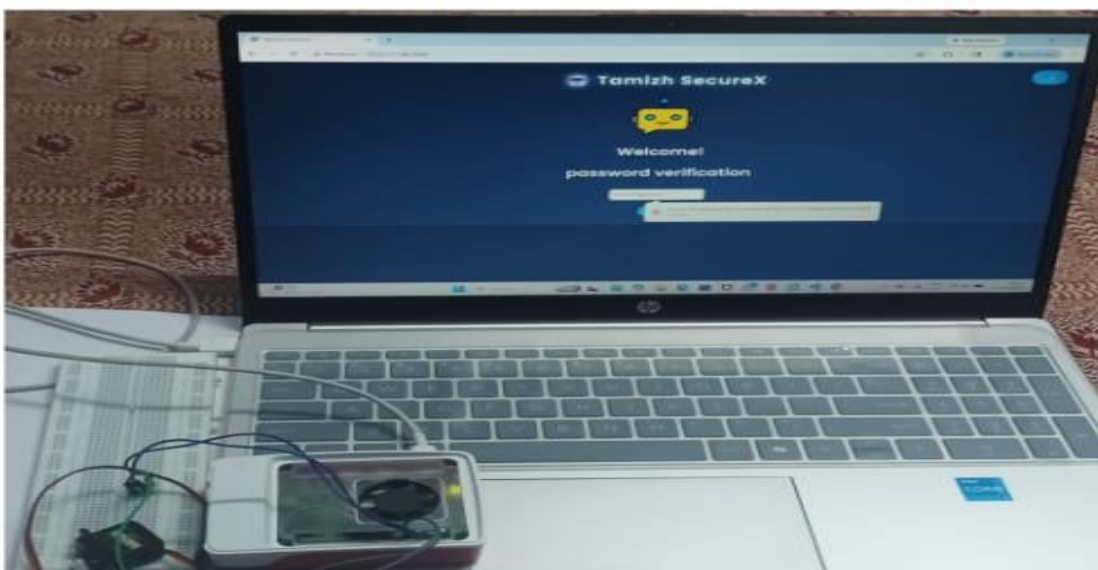
HTML

JavaScript

CSS

PYTHON

9. Results and Discussion

[illegible]



Welcome!
password verification

Next



Pattern verification!
Draw Pattern



Next

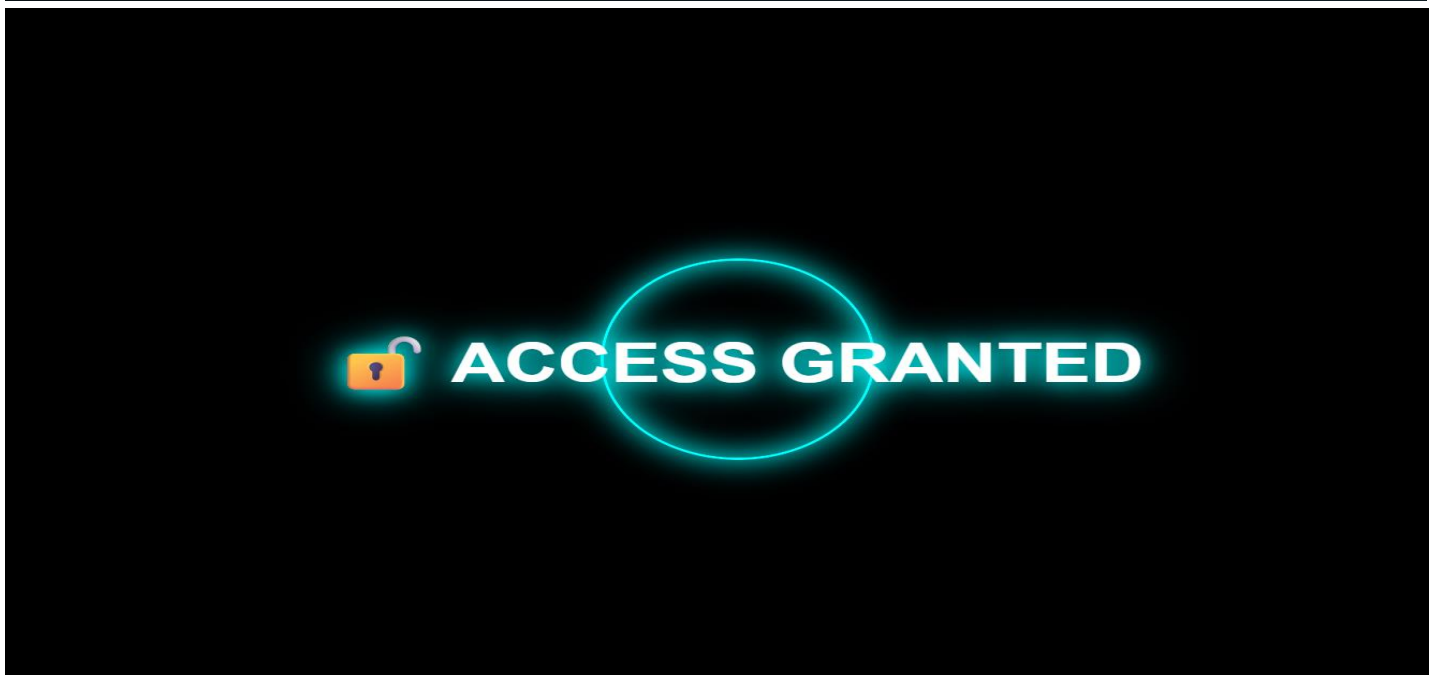
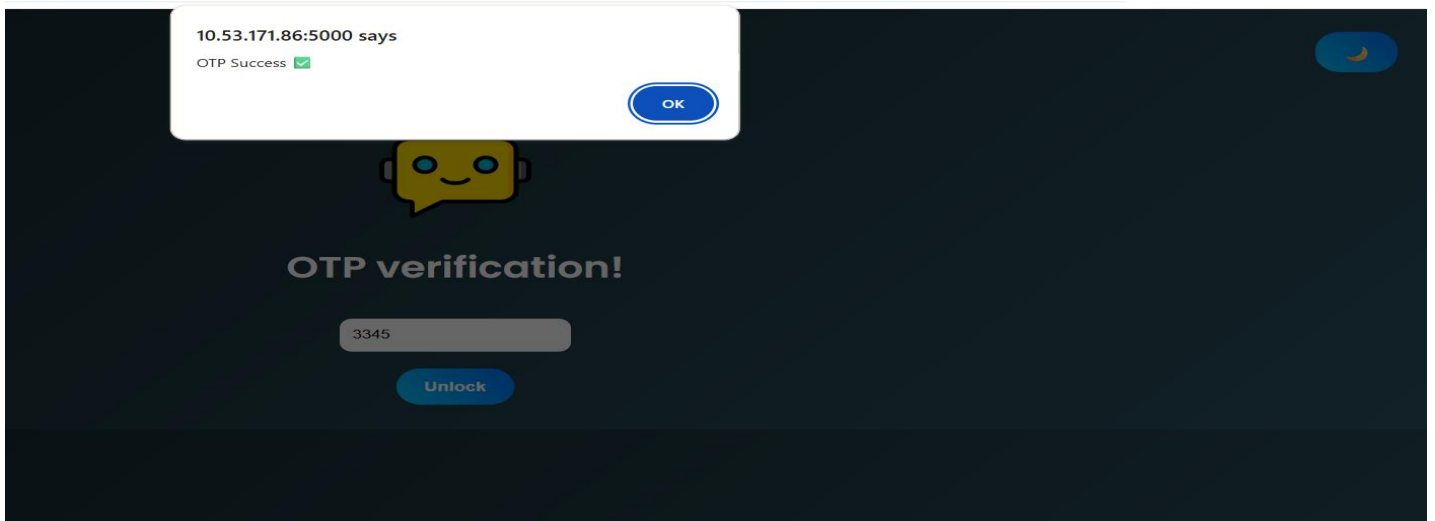
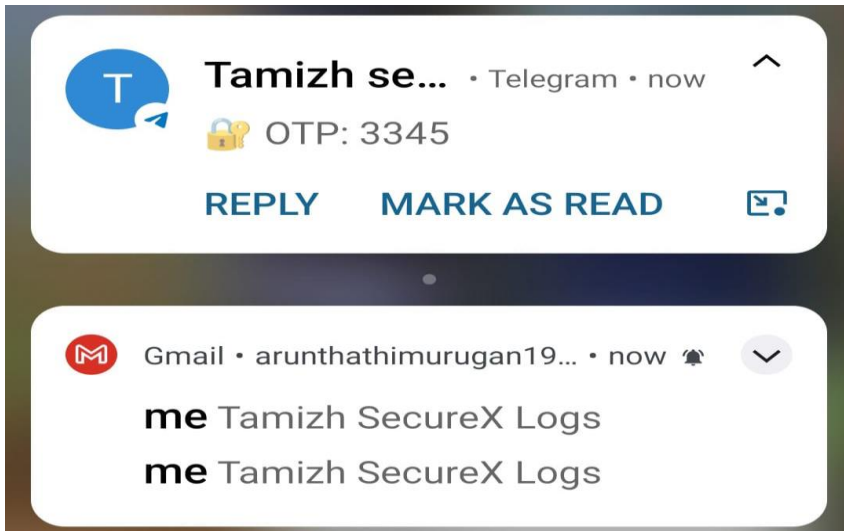


Face verification!

 10.53.171.86:5000

Face Verified  OTP Sent 

OK





Tamizh SecureX Logs

Inbox

 arunthathimuru... 12:54
to me

	A	B	C
1	29-03-2026	11:03:23	FACE FAIL
2	29-03-2026	11:31:16	UNLOCKED
3	29-03-2026	11:48:18	UNLOCKED
4	29-03-2026	11:51:16	UNLOCKED
5	29-03-2026	12:05:54	UNLOCKED
6	29-03-2026	12:12:37	FACE FAIL
7	29-03-2026	12:13:27	UNLOCKED
8	29-03-2026	22:04:37	UNLOCKED
9	29-03-2026	22:41:26	WRONG OTP
10	29-03-2026	22:42:00	WRONG OTP
11	29-03-2026	22:42:39	WRONG OTP
12	29-03-2026	23:12:40	FACE FAIL
13	30/03/2026	00:03:03	UNLOCKED



logs.csv

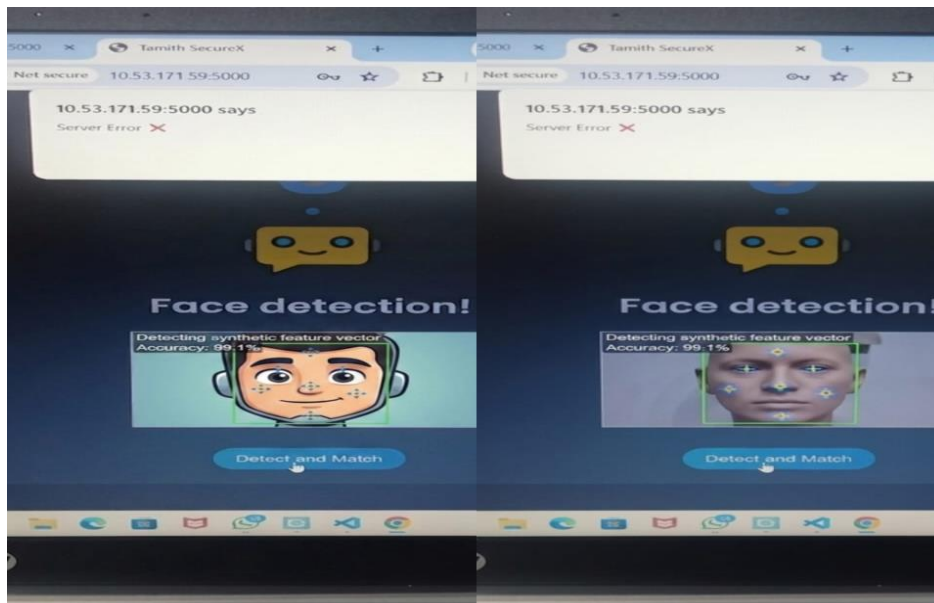
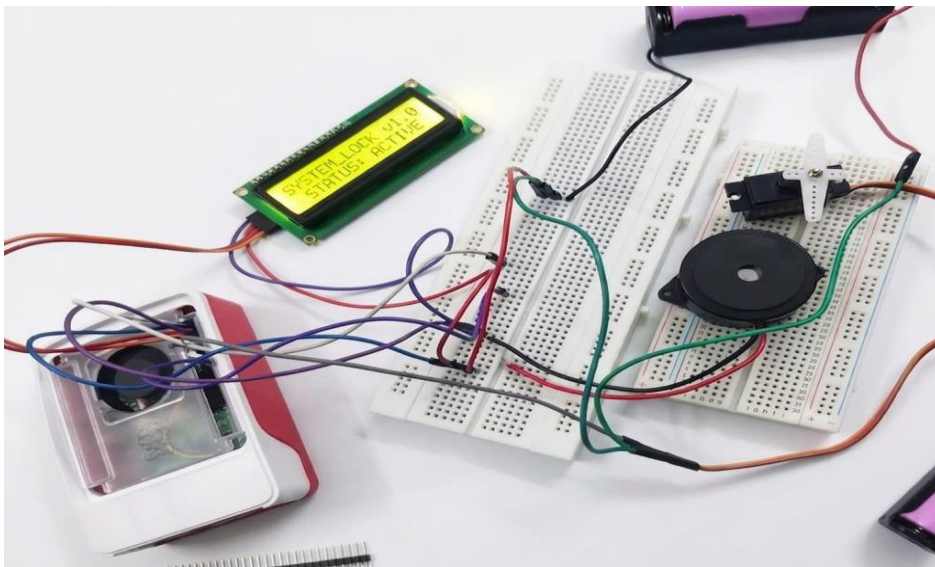


logs (11)

	A	B	C	D	E	F	G
1	29/03/2026	11:03:23	FACE FAIL				
2	3/29/2026	11:31:16	UNLOCKED				
3	3/29/2026	11:48:18	UNLOCKED				
4	3/29/2026	11:51:16	UNLOCKED				
5	3/29/2026	12:05:54	UNLOCKED				
6	3/29/2026	12:12:37	FACE FAIL				
7	3/29/2026	12:13:27	UNLOCKED				
8	3/29/2026	22:04:37	UNLOCKED				
9	3/29/2026	22:41:26	WRONG OTP				
10	3/29/2026	22:42:00	WRONG OTP				
11	3/29/2026	22:42:39	WRONG OTP				
12	3/29/2026	23:12:40	FACE FAIL				
13	3/29/2026	23:21:02	FACE FAIL				
14	3/29/2026	23:29:12	FACE FAIL				
15	3/29/2026	23:48:01	UNLOCKED				
16	3/30/2026	0:03:03	UNLOCKED				
17	3/30/2026	14:48:43	UNLOCKED				
18	3/31/2026	14:13:37	UNLOCKED				
19	3/31/2026	14:14:15	FACE FAIL				
20	3/31/2026	14:15:13	FACE FAIL				
21	3/31/2026	14:15:50	FACE FAIL				
22	3/31/2026	14:17:59	FACE FAIL				
23	3/31/2026	14:18:21	FACE FAIL				
24	3/31/2026	14:19:12	FACE FAIL				
25	3/31/2026	14:22:28	FACE FAIL				
26	3/31/2026	14:22:54	FACE FAIL				
27	3/31/2026	14:27:38	FACE FAIL				
28	3/31/2026	14:36:38	FACE FAIL				
29	3/31/2026	15:13:21	FACE FAIL				
30	3/31/2026	15:13:38	FACE FAIL				
31	3/31/2026	15:13:58	FACE FAIL				
32	3/31/2026	15:30:48	UNLOCKED				
33	3/31/2026	15:34:55	FACE FAIL				
34	3/31/2026	15:35:20	FACE FAIL				
35	3/31/2026	15:38:44	FACE FAIL				
36	3/31/2026	15:39:56	FACE FAIL				
37	3/31/2026	15:40:20	FACE FAIL				
38	3/31/2026	15:41:24	FACE FAIL				
39	3/31/2026	15:41:53	FACE FAIL				
40	3/31/2026	15:42:37	FACE FAIL				
41	3/31/2026	15:43:32	FACE FAIL				
42	3/31/2026	15:44:15	UNLOCKED				
43	3/31/2026	15:53:23	UNLOCKED				
44	3/31/2026	15:53:41	FACE FAIL				
45	3/31/2026	16:07:34	UNLOCKED				
46	3/31/2026	16:23:11	UNLOCKED				
47	3/31/2026	16:38:03	UNLOCKED				
48	3/31/2026	16:41:27	UNLOCKED				
49	3/31/2026	16:58:02	UNLOCKED				
50	3/31/2026	19:30:52	UNLOCKED				
51	3/31/2026	19:42:34	WRONG OTP				
52	3/31/2026	19:44:05	WRONG OTP				
53	3/31/2026	19:46:37	UNLOCKED				
54	3/31/2026	19:50:16	FACE FAIL				
55	3/31/2026	19:50:41	UNLOCKED				
56	3/31/2026	19:53:04	UNLOCKED				

10. Future Enhancements

- Add LCD display
- Add mobile app control
- Integrate fingerprint sensor
- Enable IoT monitoring
- Face Detection



11. Conclusion

The Password-Based Smart Lock System using Raspberry Pi has been successfully designed and implemented. The system provides a reliable and efficient method of securing access using password authentication. It eliminates the drawbacks of traditional locking systems and enhances security.

The project demonstrates how embedded systems and simple electronic components can be combined to create a smart security solution. With further improvements, the system can be upgraded into an advanced smart lock with multiple authentication methods.